

1 BDE and MDE Functions

This document describes the different *AIP* functions used for the Shop Floor Data and Machine Data Collection (BDE and MDE).

1.1 Logging on an operation

When you log on an operation, the following methods are available:

Separate logon of orders and staff	Operations and staff have to be logged on/off separately. You have to enter the staff badge number with the function Log on operation . This is a validation check only. The person must be logged on separately using the function Log on person .
Log person on with order	The OP and the person are logged on in one posting process. You only use the function Log on person if further persons are logged on to this operation.

Make this setting in the terminal configuration on the client (*Log user on with order* option).

Posting process

Select a workplace before you log on an operation. When you then call the dialog, the workplace field is already populated.

Calling the function *Log on operation*

Click the button *Log operation on*.

When the function is called, the user is guided through the input dialog.

Choose operation

Manual entry via keyboard

or

Selection via sequencing list (see the "Notes on the sequencing list")

or

Scan of bar code



With manual entry or bar code scan, the operation is not automatically searched and positioned in the sequencing list.

Choose status

Enter or select the status number that must be set when the operation is logged on.

Staff badge number

The input field is also displayed if you have selected the "separated" option for the "logon" in the terminal configuration (client). The person is not logged on with the order but is only used for validation. For this setting, go to the terminal configuration: *authorizations on the terminal > log OP on*.

Confirmation Log on operation

You use the button **Log on operation** to log on a new operation. With the OP logon, the run times are booked to the different time accounts and the quantities are posted for the OP.

When the operation is logged on, the *Workplace* field in the backlog of orders is filled with the workplace where the operation is logged on. If the operation has been planned for another workplace, the planned workplace is then overwritten. As a result, the OP is implicitly re-planned. This re-planning does not involve any further actions (e.g. update of the template, rescheduling).

Notes on the sequencing list

You can limit the number of entries in the sequencing list for the specific workplace. The list should not get too long because a long list has a negative effect on response time behavior, operability and search time (recommendation: not more than 50 entries, less with remote connections).

The same is valid for the functions *Interrupt operation*, *Finish operation* and *Partial confirmation*. Here, running operations are displayed.

Filter						
Article	Order	Operation	Target qty.	Yield	N	T
SAL-VGO011-M-9001	S3000210	0400 Cleaning	550 PCS	0 PCS		
SAR-VGO011-M-9001	S3000211	0400 Cleaning	600 PCS	0 PCS		
SAR-VTO021-K-5013	S3000515	0200 Deburring	1800 PCS	0 PCS		
SAL-FMO021-K-7036	S3000506	0200 Deburring	2200 PCS	0 PCS		
SAR-FMO021-K-7036	S3000507	0200 Deburring	2300 PCS	0 PCS		
SAR-VTO021-K-3000	S3000519	0200 Deburring	1900 PCS	0 PCS		
SAL-FMO021-K-3000	S3000508	0200 Deburring	1800 PCS	0 PCS		
SAR-FMO021-K-9001	S3000503	0200 Deburring	1400 PCS	0 PCS		
SAR-VSH021-K-5013	S3000523	0200 Deburring	1600 PCS	0 PCS		

If an operation is already logged on to the workplace, then the system enters this workplace in the input field *Operation*. If you do not want this and if the system should preset the first operation of the sequencing list in this field, then you must make the following configuration:

- Call the application [Dynamic dialog configuration - Fields](#).
- Request data for the dialog "WF_AGL". If it is not the logon dialog A_AN or A_P_AN, then identify the dialog.
- Select the field "ANR".

- Click the *Edit* button (note: this is only possible with terminal-specific or terminal group-specific dialogs. It is not possible with AIPDEF 0 dialogs).
- Enter the value "SETVALUE" (without quotation marks) in one of the field attributes 1 to 8.
- Leave the field *Default* empty.
- Save your changes.
- Change to the application [Dynamic dialog configuration](#).
- Restart the dialogs and then restart the terminals.

Displaying notes and texts

Via configuration in the application [Order types](#), you can use the option *Show OP info when logging on OP* to specify for a specific order type that after a successful order logon on the AIP either

- the notes on the operation with an active option *Visible on the terminal*

or

- the long texts of operations

are automatically displayed. The "OP info" dialog is then opened with the respective active page.

Note: To show the information, the AIP must be connected online to the server.

1.2 Logging an operation off

You use the button *Log off operation* on operation level to log off an operation. The posting of run times and quantities is then finished for the OP. Once logged off, you cannot log on the OP again.

Posting process

Select the workplace and the operation that must be logged off in the main view. When the dialog is called, these fields are then preassigned and cannot be changed.

Calling the function *Log operation off*

Click the *Log operation off* button.

When the function is called, the user is guided through the input dialog.

Yield

Enter the yield quantity that must be posted.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). Enter the produced scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Deviation reason

You can activate an overdelivery or underdelivery check for operations or persons. When the operation is logged off, the system then checks if the quantity recorded has exceeded the overdelivery limit or falls short of the underdelivery limit. If the check has a positive result, the quantities posted are rejected with the error message "Overdelivery" or "Underdelivery".

If only a warning is activated for the check, the user can force the system to accept the overdelivery or underdelivery by entering a deviation reason.

Status

Enter or select the number of the workplace/machine status that is set after the operation is logged off.

Estimated duration

Optionally enter the estimated duration of the selected workplace/machine status in minutes.

Note: Only make an entry here if the status is a downtime status (status <> production).

Comment

Enter an optional comment for the status in this field that is also displayed in the machine history.

Staff badge number

The person entered must be authorized to log off the operation. The setting is made in the HR master data. Go to: Shop floor data > BDE authorizations > Lop OP off.

Confirming Log off operation

The operation is logged off, once the dialog has been confirmed. You cannot log the operation on again.

1.3 Interrupting an operation

Click the button *Interrupt operation* to call this function. You use this function to stop collecting times and quantities for an order. The reasons for the interruption can be a quantity upload, a shift change or an interruption of the production for technical reasons.

The process of interrupting an operation and the layout of the input dialog are identical to the ones of the operation logoff. The difference is that you can log on an interrupted operation at any time.

1.4 Uploading a part quantity for an operation

You can use this function to upload a part quantity for the order without interrupting or finishing the running OP. The quantities are booked for the relevant OP and the person that makes the posting. Click the button *Confirm partially* to call this function.

Posting process

Select the workplace and the operation from the list. When you then call the dialog, the system preassigns these fields and the values cannot be changed.

Calling the function *Confirm partially*

Click the button *Confirm partially*.

When the function is called, the user is guided through the input dialog.

Yield

Enter the yield quantity that you want to post.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). You can enter scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Deviation reason

You can activate an overdelivery or underdelivery check for operations or persons. The system then checks if the overdelivery limit is exceeded with the quantity posted. If the check has a positive result, the quantity posted is rejected with the error message "Overdelivery".

If only a warning is activated for the check, the user can force the system to accept the overdelivery by entering a deviation reason.

Staff badge number

The person entered must be authorized to upload part quantities for the operation.

1.5 Log person on

Persons are logged on to or off from a workplace. The logging of persons is therefore made on the workplace level.

You can only log on a person, if an operation has been logged on to the workplace.



You can only use this function to log on persons with single workplaces or production orders.

With group workplaces or overhead cost operations, the persons are logged on via the combined logon of order and persons when the operation is logged on. It must be guaranteed

here that a staff badge number can be entered during the posting process.

Posting process

Select the workplace where you want to log the person on or off and click the button *Staff logging*. The system displays a list of the *Registered persons* that are already logged on to this workplace. Close this view by clicking *Close information*.

Calling the function *Log person on*

Click the button *Log person on*. A dialog including only one dialog step opens.

Staff badge number

To identify the person, also enter the staff badge number here.

Other notes

Via customization, you can optionally enter an operator position/function or a premium indicator. You specify in the course of the customization how the value is entered: only a field with a selection list or a separate workflow.

You can configure that an advance logon for the next shift is possible within a configurable time before the start of the next shift. Configure this time in the terminal configuration (tab: *MF functions* > *Waiting period for advance logon of staff*). But if the last operation is logged off before the end of the current shift, all advance logons are deleted. And you cannot log off a person that is logged on in advance.

1.6 Log person off

Persons are logged on to or off from a workplace. The logging of persons is therefore made on the workplace level. You can only log off a person that has been logged on to the workplace.



You can only use the function to log off persons with single workplaces or production orders. With group workplaces or overhead cost operations, the persons are logged off via the combined logoff of order and persons when the operation is interrupted or logged off.

Posting process

Select the workplace where you want to log the person on or off and click the button *Staff logging*. The system displays a list of the *Registered persons* that are already logged on to this workplace. Close this view by clicking *Close information*.

Calling the function *Log person off*

Click the button *Log person off*. A dialog including only one dialog step opens.

Yield

Enter the yield quantity that you want to post.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). You can enter scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Staff badge number

To identify the person, also enter the staff badge number here.

1.7 Log off everyone

You can use this function to log off all persons in one posting that are logged on to a machine.

You assign the authorization to use this function in the HR master data. Activate the following option in tab *Shop floor data > BDE authorizations > Log all staff off*.

Posting process

The posting process is in general identical to the dialog *Log person off*. The only difference is that you cannot record quantities when using the dialog *Log off everyone*.

1.8 Change workplace/machine status

You can use this function to assign a new status to a workplace/machine. You might need this function during setup of the workplace or in case of a malfunction, for example.

A status should be entered, when the AIP automatically identifies a malfunction and the status changes to "not assigned" (also see "Monitoring of operating signals and cycle time").

You can configure that only persons can perform a status change that are already logged on. Enable the following option in the HR master: tab *Shop floor data > BDE authorizations > Change only if person is logged on*.

Posting process

Select the workplace where you want to change the status and click the button *Change status*.

Status

Enter or select the status that you want to set for the workplace.



With manual entry or bar code scan, the status is not automatically searched and positioned in the selection list.

Estimated duration

You can enter an estimated time in minutes that the new status will probably take. This field should only be filled for downtime statuses.

Comment

You can enter a comment for the status. This comment can be displayed in the machine history.



If the status lasts on and shifts change, then the comment is only assigned to the MDE log record of the shift before shift change.

Staff badge number

Enter the badge number of the person that changes the status. The number is required for the validation check. The person must be authorized to change the status.

Confirmation of the dialog

Once the dialog has been posted successfully, the new status is activated for the machine.

Other notes

It is possible to change from one status to another, unless the automatic status monitoring has been activated for the workplace/machine (workplace/resource configuration > *tab: MDE configuration* > *automatic monitoring: cyclic or operating signal*) and the workplace is in status “production”.

If you do not want to show a machine status in the status list (any more), then you must disable the option *Status manually at terminal* in the machine status assignment.

Hierarchical statuses

You can build a status hierarchy as of HYDRA-MDE 7.2. In the status assignment, you store the status number of the direct superordinate status for a lower-level status. All statuses that you cannot assign on the terminal and that are only used to show the hierarchy are called “hierarchy level”.

Hierarchy levels are displayed in blue font in the status list.

If you select a hierarchy level and double-click/touch the  button, the list of “lower-level” statuses opens. To get to the next higher hierarchy level, click/touch .

You can only select a status in the status list that can be assigned manually. The following error message is shown, if a machine status change on a hierarchy level cannot be assigned.

1.9 Lock/unlock production status



The button *Lock production status* or *Unlock production status* is only active for a workplace if the current status is not "production" and if one of the following conditions is fulfilled:

- The PCC receiving the machine data of the workplace runs in embedded mode and the workplace is assigned to a terminal configured with "MDE operation".
- The PCC receiving the machine data of the workplace runs in stand-alone mode and the INI configuration [MDE-NOTIFICATION](#) is activated for the workplace.

If the function *Lock production status* is active, the terminal cannot automatically change to status "production" when the terminal identifies signals (pulses or operating signal).

If the production status is locked, this means:

- With an active lock, the automatic change to status "production" is not possible and the status currently set is kept despite machine pulses (e.g. "setup").
- If cycle signals are recorded during the lock via counter inputs configured as "yield", then the relevant quantities are booked according to the [Configuration](#) of *Posting during prod. lock*:
 - as yield,
 - as scrap
 - or not at all.
- Click the button *Unlock production status* to remove the lock.

The production status can also be locked/unlocked when a specified workplace/machine status is set. You can configure the behavior in the configuration [Status assignment](#) for each status that is not "production" (go to: *Master data > Workplaces/machines > Status assignment*).

Logging of the production lock

The manual lock or unlock of the production status is logged as event on the server and can be shown in the [Machine history](#). Note: the logging of the event is only performed if the status is manually locked or unlocked.

But if the production status is locked or unlocked via status change, then this is not documented explicitly as an event and is therefore not shown in the machine history.

If the production status is locked for a machine and the terminal software is restarted, then the production lock is automatically removed after this restart. The changed production lock is not logged.

Authorization check when setting the production lock manually

You can configure that only if a relevant authorization is available, the person is allowed to manually (explicitly) set or remove the production lock.

To do so, activate the dynamic dialog M_PSPERRE. If this dialog is activated, you must enter the staff badge number.

If the dialog is activated, the dialog is displayed when the operator clicks the button *Lock production status* or *Unlock production status*. Once the badge number has been entered, the system checks whether this person is authorized to lock/unlock the production status. The system checks, if the option *Change of production lock* in the HR master data (tab *Shop floor data*) is enabled.

If the terminal is OFFLINE, the terminal configuration specifies the behavior (option *Checking required*).

1.10 Change target cycle

The target cycle is the default value that is checked in case of a machine monitoring based on cycle time. You can use this dialog to change the target cycle for the machine.

Posting process

Select the workplace where you want to change the target cycle. Click the button *Change target cycle*.

New target cycle

Specify the new target cycle in seconds/cycle

Staff badge number

The entry of the staff badge number is optional. If configured accordingly, a validation check is performed for the number. (*HR master data > tab: Shop floor data > Change of cycle/partitioning*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new target cycle is activated for the machine.

1.11 Change partitioning

The partitioning (also called cavity) specifies the number of parts produced per machine cycle (clock). You can use this dialog to change the partitioning. The automatic collection of quantities is then based on this partitioning.

Posting process

Select the workplace where you want to change the partitioning. Then click the button *Change partitioning*.

Partitioning

Enter the new partitioning.

Staff badge number

You can optionally enter the staff badge number. If configured in the HR master data, a validation check is performed for this number (tab: *Shop floor data > BDE authorizations > Change cycle/partitioning*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new partitioning is used for all running operations at the machine.

1.12 Change target quantity

Use this dialog to change the target quantity based on operations (primary quantity unit).

Posting process

Select the operation where you want to change the target quantity and click the button *Change target quantity*.

New target quantity

Enter the new target quantity that you want to store for the operation.

Staff badge number

You can optionally enter the staff badge number. If configured in the HR master data, a validation check is performed for this number (tab: *Shop floor data > BDE authorizations > Change target quantity*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new target quantity is stored for the operation.

1.13 Information on operations (OP info)

General information

Use the button  on operation level to call the OP info dialog¹. Select the required operation. A dialog opens. The dialog includes several pages that are organized in tabs. The information is only requested from the database when you call the relevant page.

¹ This function is also available at other places, e.g. in the *Log operation on* dialog.



Order S3000209 Operation 0200 Deburring

Description
Notes
SF-Comments
Documents
Tools, Resources
Components
Progress
Resource performance acc.

Operation	S30002090200	Target qty.	1200	PCS
Article	SAR-VPO011-M-9005	Yield	660	PCS
Article Designation	Mirror housing right, bla	Scrap	0	PCS
Remark 1		Completion	55%	
Remark 2				
Planned duration	06:40			

FURTHER INFORMATION:
- EVERY 750 PIECE: QC ACCORDING TO FORMULAR 823273
- PACKING / TRANSPORT STD-CONTAINER NO 4

Close OP information


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The dialog includes the following tabs:

Description

Current information on the current operation is displayed.

- Operation (MES order number)
- Article
- Article designation
- Remark 1
- Remark 2
- Planned duration
- Target quantity (of the operation)
- Yield

- Scrap
- Completion (in %)

At the bottom, the long text assigned to the operation is displayed.

Notes

The tab *Notes* displays the notes entered on the client (usually via the *Graphic planning board*) if the option *Display on terminal* is enabled.

The table shows all notes of the operation that are configured to be displayed on the terminal. The list is sorted by the editing date. The most recent note is on top and is displayed when you call the view.

If you click/touch a note of the list, the complete text is displayed in the field below.

1.13.1 SF comments

The tag *SF comments* displays the comments recorded during the Shop Floor Data Collection (BDE) or you can record new comments.

The comments recorded are displayed in the *Order information* dialog on the client or can be forwarded via escalation management, e.g. by e-mail.



If an SF comment is recorded for an operation that has been merged on the MOC, then the SF comment is only relevant for this merged operation. The SF comment is not transferred to the single operations.

You cannot record SF comments for the single operations because the single operations are not displayed on the AIP.

With split operations, the SF comment is only stored for the split operation where the SF comment has been recorded. The SF comment is not transferred to the split master.

1.13.2 Documents

You can display documents, graphics or other files on the AIP, which are listed in the table in tab *Documents*.

Select an entry in the list and touch/click the button *Open document*. The file is downloaded to the AIP and displayed using an internal viewer or an external application (according to the file extension).

Internal viewer

Supported formats for the internal viewer: txt, ini, avi, tif, tiff, jpg, jpeg, dcx, eps, ico, pcx, pcc, png, ppm, pgm, pbm, tga, vst, afi, wmf, emf, bmp.

Supported formats for external HTML viewer: htm, gif, wmv, mpg,

External applications

You must install external applications if other file formats (file extensions) than the ones mentioned above are used (e.g. PDF files). The customer is responsible for the installation.

http links as document references

You can also pass http links to a browser for display without having to download a file beforehand. Use a path with the "http" schema (paths are configured on the client via *System administration > System settings > Paths*).

These links are displayed using the internal HTML viewer provided by the *AIP*. The file extension does not affect the selection of the viewer.

Also the default browser configured in Windows can be used for the display. To do so, configure the following option in the "hytnrcfg.ini" file:

```
[Terminal->USR 0]
```

```
HTTPBrowser=standard
```

This setting is not recommended for an *AIP* with touch screen, because the operation of a browser can lead to problems.

1.13.3 Tools, Resources

The tab *Tools, Resources* displays the production resources and tools required for an operation.

Note: Documents are displayed separately in tab *Documents*.

1.13.4 Components

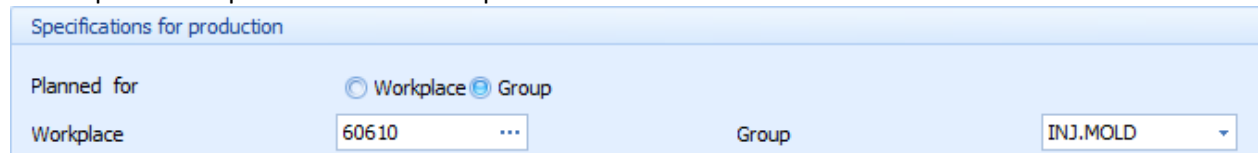
The tab *Components* displays the material components required for the operation.

1.13.5 Progress

The tab *Progress* displays information on the status of the different operations of the order that includes also the currently the selected operation. The below-mentioned data is shown:

- Order
- Operation
- Operation designation

- Color of the status according to the control indicator of the status:
S – (no color), V – gray, L – light green, U – yellow, E – green
- Status (text) of the operation
- Target quantity of the operation
- Quantity unit of the operation
- Yield that has been posted so far for the operation
- Quantity unit of the operation
- Workplace that is assigned to the operation according to the order management. It does not matter if the operation is planned for this workplace or not.



Specifications for production

Planned for ☐ Workplace ☒ Group

Workplace ... Group

- Group the workplace is assigned to according to the workplace configuration.

1.13.6 Resource performance accounts

In tab *Resource performance accounts*, the following information is displayed for the resource performance accounts (RPA) 1 to 11 of the current operation:

- RPA abbrev.
- RPA designation
- Posted duration in hours:minutes
- Total duration = total of all times of RPA 1...11

The durations are displayed in a graphic (to the right).

1.14 Machine information (machine info)

The machine information dialog provides the views and functions listed below.

1.14.1 Description

Machine 33021 Manual Workplace Deburring, Cleaning

Description		Registered persons	Status log
Workplace/Machine	33021	Status	PRODUCTION
Short description	WPL02	Status since	21600 02/28/2010
Group	FINISH	Duration [hh:min]	0051:07:18
Divisor	1	Yield	833PCS
Target cycle [sec/cycle]	20	Scrap	4PCS
		Cycles	0

Close information
07/12/12 18:29:40

The tab *Description* shows information on the machine/workplace

Workplace/machine

Number of the workplace/machine according to configuration

Short name

Short name of the workplace/machine according to configuration

Group

Group the workplace/machine is assigned to according to configuration.

Partitioning

The current partitioning and the incoming cycles are used to calculate the number of parts per cycle posted for the workplace/machine.

Target cycle

Current target cycle used to monitor the workplace/machine.

Status

Current status

Status since

Point in time that specifies the beginning of the current status.

Duration

The duration specifies how long the current status has been available.

Yield

If the workplace has been configured as MDE workplace, this field shows the yield that has been posted in the current shift up to now.

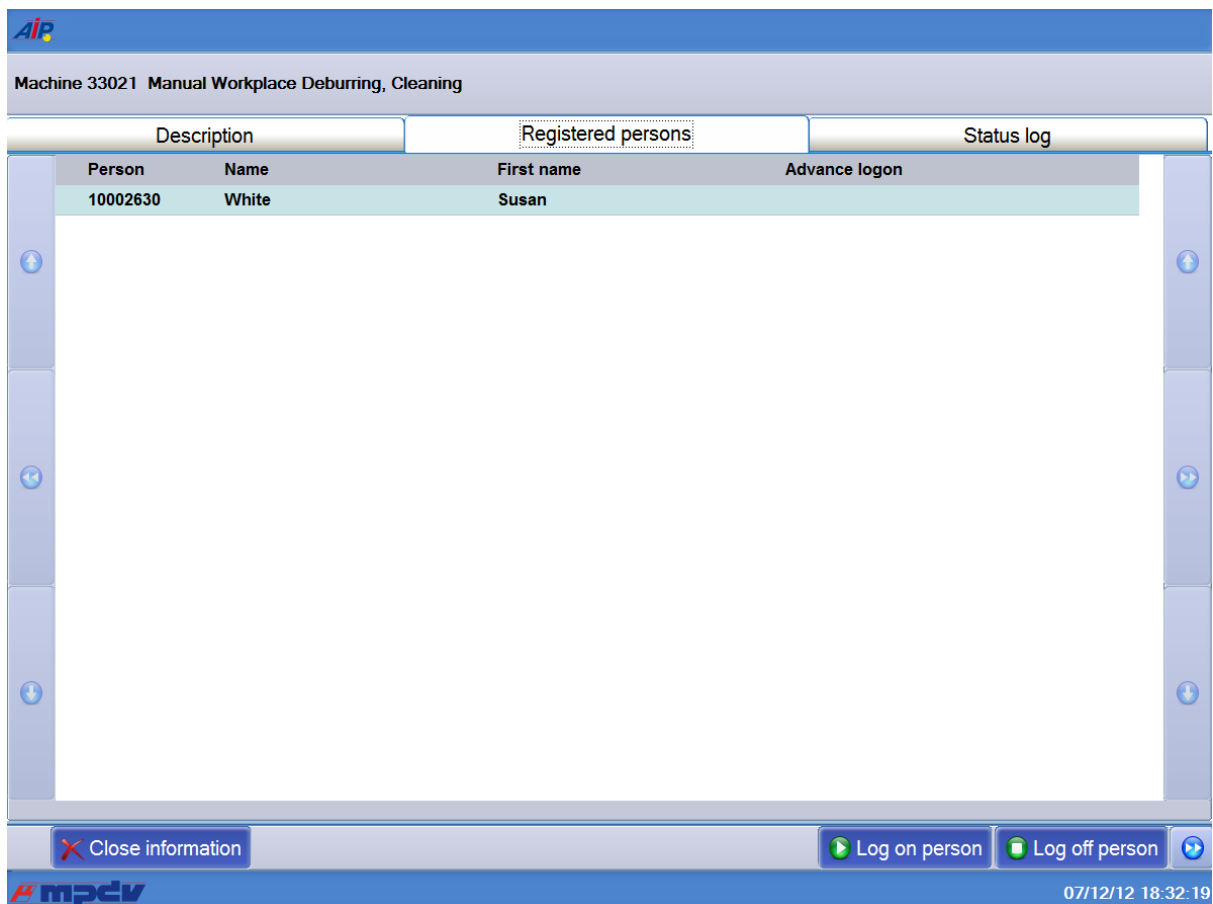
Scrap

If the workplace has been configured as MDE workplace, this field shows the scrap that has been posted in the current shift up to now.

Cycles

If the workplace has been configured as MDE workplace, this field shows the cycles that have been recorded in the current shift up to now.

1.14.2 Registered persons



Machine 33021 Manual Workplace Deburring, Cleaning

Description		Registered persons	Status log
Person	Name	First name	Advance logon
10002630	White	Susan	

Close information Log on person Log off person

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This overview shows the staff currently logged on to the workplace.

Use the following buttons to perform postings for staff:

- [Log on person](#),
- [Log off person](#) or
- [Log off everyone](#) (on the next button page)

Click *Close information* to get back to the main view.

1.14.3 Status log

Machine 33021 Manual Workplace Deburring, Cleaning

Description Registered persons Status log

Shift date 03.01.2010 Shift 1

☐ Display statuses without reason only

Event	Date	Time	Duration	Order	OP	Person	Name
Production	16.03.2010	06:00:00	04:10				Reißfelder
OP logged on	16.03.2010	06:00:00	08:00	000001177424	0002	9282	Reißfelder
Person logged on	16.03.2010	06:00:00	08:00	000001177424	0002	9282	Reißfelder
Tool defect	16.03.2010	10:10:00	00:13			1256	Reißfelder
Production	16.03.2010	10:23:20	01:01				STEINBEISSER GEORG
General disturbance	16.03.2010	11:25:00	00:16				STEINBEISSER GEORG
Production	16.03.2010	11:41:40	02:18				STEINBEISSER GEORG
Person logged on	16.03.2010	14:00:00	00:00	000001177424	0002	4232	Reißfelder
Production	16.03.2010	14:00:00	00:18				STEINBEISSER GEORG
OP logged on	16.03.2010	14:00:00	08:00	000001177424	0002	4232	Reißfelder
OP interrupted	16.03.2010	14:00:00	00:00	000001177424	0002	9282	Reißfelder
Person logged off	16.03.2010	14:00:00	00:00	000001177424	0002	9282	Reißfelder
Stamp change	16.03.2010	14:18:24	00:05			9578	STEINBEISSER GEORG
Production	16.03.2010	14:23:57	00:03				STEINBEISSER GEORG

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Change Status

Close information

07/12/12 18:33:06

At first, the table shows the following events of the current shift:

- All machine/workplace statuses
- Production lock (set manually, removed manually)

- Postings relating to orders (OP logged on, OP interrupted, OP logged off)
- Postings relating to staff (person logged on, person logged off)

Use the buttons  and  to scroll back and forth shift by shift. You can configure for the specific terminals how many shifts you can scroll back and forth. Use the following option in the `hytnrcfg.ini` (3 shifts by default):

```
[MDE->Options 0]  
MSTAT.SKNRRANGE=3
```

In the above example, the screen shows shift 1 if you click  in shift 3. Scrolling back has the same behavior.

Subsequent assignment of reasons

If the machine monitoring function is active, the status switches to the “not assigned” status in case of an absence of signals. When the malfunction has been removed, the machine immediately switches into the production status. The operator cannot enter a reason for this status.

In the status log, you can now list these statuses without reason and subsequently assign a reason to the statuses.



Statuses without reason are statuses that the system has automatically set from “not assigned” to “production” in the course of an active machine monitoring. The production indicator “general disturbance” is assigned to this status and the relevant time is posted for this status.

If an operator manually sets a status, this status usually has a reason assigned, also if the operator uses the production indicator “general disturbance”.

If you enable the checkbox *Display statuses without reason only*, you can filter the list and directly show the statuses without reason only (i.e. automatically assigned reasons). These statuses can then be changed. Note: When you have changed the selection, you must reload the list using the green arrow button!

The *Change status* button gets active when you select a status without reason. You cannot change statuses set manually by the user. The button is grayed out (disabled).

If you click the *Change status* button, you can subsequently assign a reason to this status.

Note

Restriction: Only the MDE data (log records) is changed when you subsequently assign a reason. Postings of other objects (e.g. BDE log records) are not changed.

You cannot subsequently assign reasons to "Short-term disturbances/malfunctions".

1.15 Merged operations

General information

You use these functions to create or to end "**merged operations**" (MOP) on the AIP terminal. A merged operation includes a group of single operations. The system records the time for the merged OP and proportionally distributes the time to the single OPs. Each person can manage a maximum of one merged operation.

You require a license to use the functions for merged operations on the AIP.

To merge operations, you must click the button *Log MOP on* in the workplace screen. Merged operations are logged off and interrupted like single operations using the *Interrupt operation* or *Log operation off* buttons.



With merged operations created on the terminal, you can only perform logon/logoff/interrupt postings on the terminal. The same also applies to the single operations included in a merged operation. Postings on the MOC are not possible.

Also note that there are some restrictions for merged operations that are described in the documentation [MBL_CollectiveOperationProcessing.pdf](#). Not all postings are possible that are possible with normal operations.

1.15.1 Log merged operation on

You can combine up to 20 operations to form one merged operation. On the *AIP*, only one operation (the new merged operation) is then displayed in the list of running operations..

Posting process

Select a workplace, before you log on a merged operation. When you then call the dialog, the workplace field is already populated.

Calling the function *Log MOP on*

Click the *Log MOP on* button in the workplace dialog. The user is navigated through the dialog.

Choose status

Enter the status number. Optionally, you can also enter the estimated duration and a comment.

Choose operation

Select all single operations from the list that you want to integrate in the merged operation. Manually select the operations in the list as usual.

When you have selected the single operations, click the button *Add operation*. All selected operations are displayed in green in the list:

Logon merged operations

1 Choose status 2 Choose operation

Filter

Article	Order	OP	OP Designation	Target qty.	Yield
SAL-VGO011-M-9001	S3000210	0400	Cleaning	550	0
SAR-VGO011-M-9001	S3000211	0400	Cleaning	600	0
SAR-VTO021-K-5013	S3000515	0200	Deburring	1800	0
SAL-FMO021-K-7036	S3000506	0200	Deburring	2200	0
SAR-FMO021-K-7036	S3000507	0200	Deburring	2300	0
SAR-VTO021-K-3000	S3000519	0200	Deburring	1900	0
SAL-FMO021-K-3000	S3000508	0200	Deburring	1800	0
SAR-FMO021-K-9001	S3000503	0200	Deburring	1400	0
SAR-VSH021-K-5013	S3000523	0200	Deburring	1600	0

Operation No.

Staff badge number

31.07.14 11:02:19

Staff badge number

Enter the staff badge number here. With merged operations, the person is always logged on with the operation, i.e. a separate staff logon is not possible.

Confirmation via *Start merged operation*

The specified operations are immediately logged on. If the terminal is ONLINE, the system makes a validation check for all entries.

When the merged operation is successfully logged on, an entry in the order overview is generated. This entry includes the characters "SAM-" for merged operation and the badge number (e.g. **0160**). The merged operation of the example would therefore be: **SAM-0160**.

1.15.2 Interrupt/log merged operation off

You interrupt or log off a merged operation like single operations. Select an operation named **SAM-
<badge number>** from the list of running operations.

Posting process

Select the workplace and the merged operation that you want to log off.

Calling the function

Click the button *Interrupt operation or Log operation off*.

Log merged operation off

Enter the badge number of the person who performs the logoff.

Choose status

Enter the status number. Optionally, you can also enter the estimated duration and a comment.

Choose operation

If the entry of quantities is configured for merged operations in the terminal configuration, a list of the different operations is displayed. To upload part quantities, you can select the relevant single operation, enter the required quantity and click the button *Partial confirmation*. The system does not immediately upload the quantities entered here. The quantities are only uploaded when you confirm the dialog by clicking the button *Interrupt/Log off merged operation*. If the dialog is canceled (ESC key or *Cancel* button), the dialog is closed and no data is uploaded. The partial confirmation is discarded.

Confirmation of the dialog *Log off/Interrupt merged operation*

The merged operation is logged off or interrupted when you confirm the dialog.

If the order is properly logged off, the merged operation is unmerged. After interruption, the single operations are available again in the sequencing list.

The quantities are posted according to the configuration and using the part quantities uploaded.

Further notes

- If you do not enter a quantity with the function *Interrupt merged operation*, then no quantities are posted. With the function *Log merged operation off*, the target quantity is booked as yield quantity.
- If you interrupt a merged operation, the system interrupts all single operations assigned to the merged operation. As a result, the sequencing list does not include the merged operation after interruption, but the single operations. If you want to log on a merged operation again, you must again select or assign the single operations.